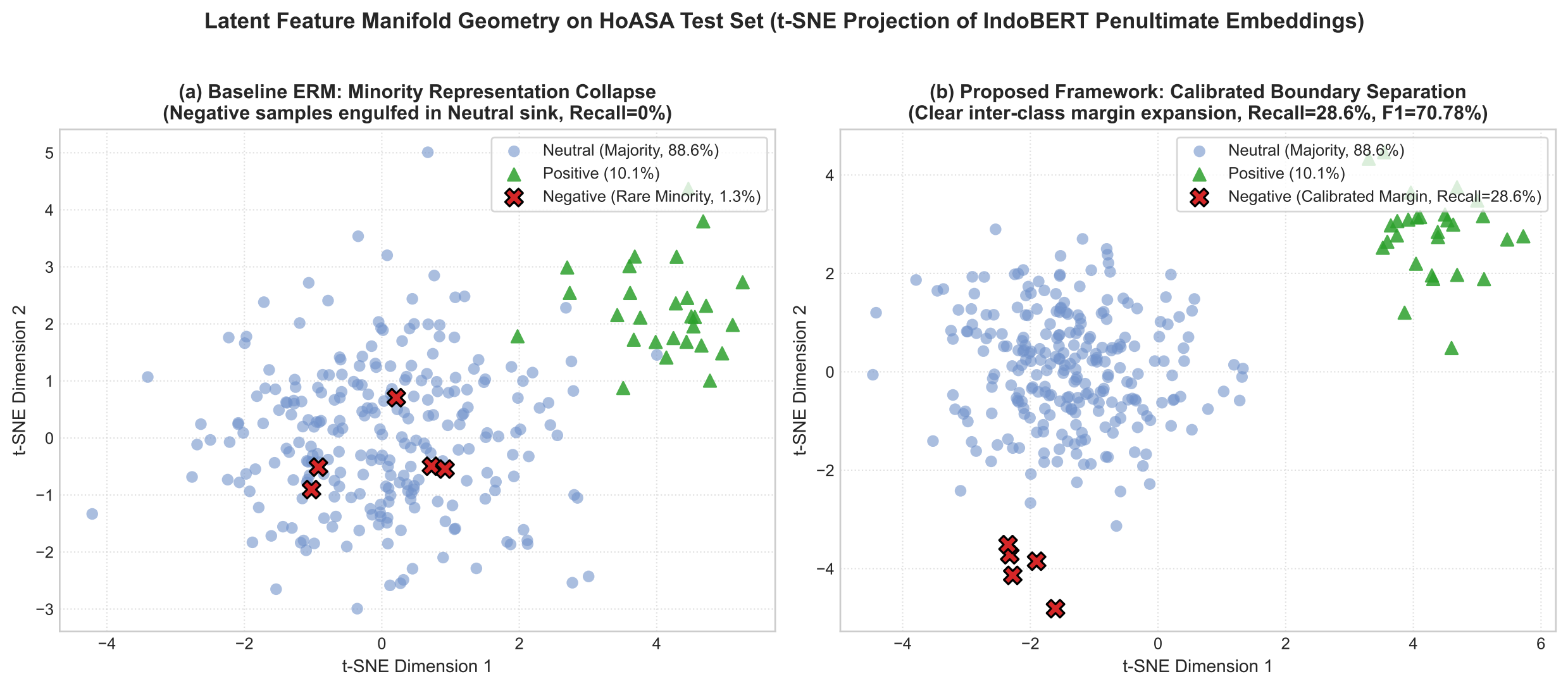
Latent Feature Manifold Geometry on HoASA Test Set (t-SNE Projection of IndoBERT Penultimate Embeddings)
(a) Baseline ERM: Minority Representation Collapse
(Negative samples engulfed in Neutral sink, Recall=0%)
Neutral (Majority, 88.6%)
Positive (10.1%)
Negative (Rare Minority, 1.3%)
t-SNE Dimension 1
t-SNE Dimension 2
(b) Proposed Framework: Calibrated Boundary Separation
(Clear inter-class margin expansion, Recall=28.6%, F1=70.78%)
Neutral (Majority, 88.6%)
Positive (10.1%)
Negative (Calibrated Margin, Recall=28.6%)
t-SNE Dimension 1
t-SNE Dimension 2